

Dissection Lab Practical CRAM

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Living Earth

Taxonomy

Domain
Kingdom
Phylum
Class
Order
Family
Genus
Species

Scientific Names:
Genus species

Kingdom: ANIMALIA

- Eukaryotic
- Heterotrophic
- Lack Cell Wall
- Multicellular
- 9 phyla

Body Regions and Symmetry

4 regions:

1. Anterior - head
2. Posterior - tail
3. Ventral - lower
4. Dorsal - upper (can be confused with the term “back”)

3 types of symmetry

1. Asymmetrical
2. Radial
3. bilateral

Body cavities and coeloms

3 embryological tissue layers:

1. Ectoderm - epidermis and nervous system
2. Mesoderm - skeletal, muscular, excretory, circulatory, and reproductive systems
3. Endoderm - entire digestive tract (liver and pancreas) and respiratory system

The term “coelomate” refers to whether the animal has a body cavity. There are three kinds of coelomates.

1. Acoelomate - no body cavity
2. Pseudocoelomate - false cavity (due to structure)
3. Coelomate - true body cavity (protostomic or deuterostomic)

Phylum Chordata

1. Notochord is replaced by a backbone
2. Nerve Cord forms a brain at anterior end
3. Pharyngeal Slits formed during embryonic stages

Chordata isn't comprised of JUST vertebrates, made up of 3 Subphyla:

- Urochordata (invertebrate)
- Cephalochordata (invertebrate)
- Vertebrata (8 classes of vertebrates)

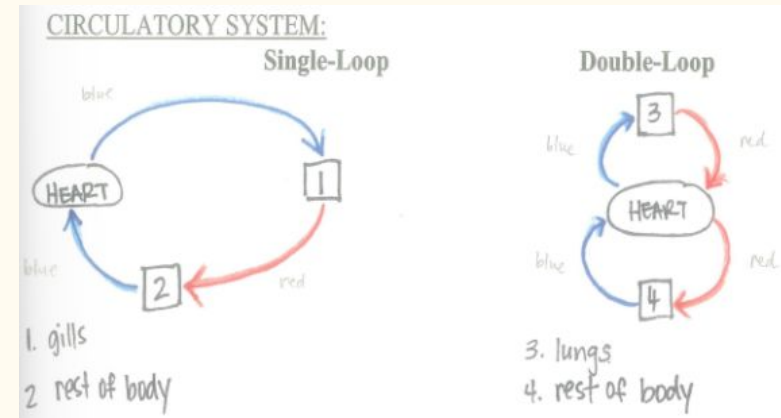
Vertebrae Reproduction

1. Oviparous - egg laying
2. Ovoviviparous - eggs develop in body until hatched
3. Viviparous - young develops inside female until giving birth to

Vertebrae Circulation

Single Loop - blood is pumped to the gills where it is oxygenated, distributed, then sent back to the heart for repumping

Double Loop - Deoxygenated blood is pumped to the lungs, oxygenated, brought back to the heart, distributed, then returns to the heart once deoxygenated



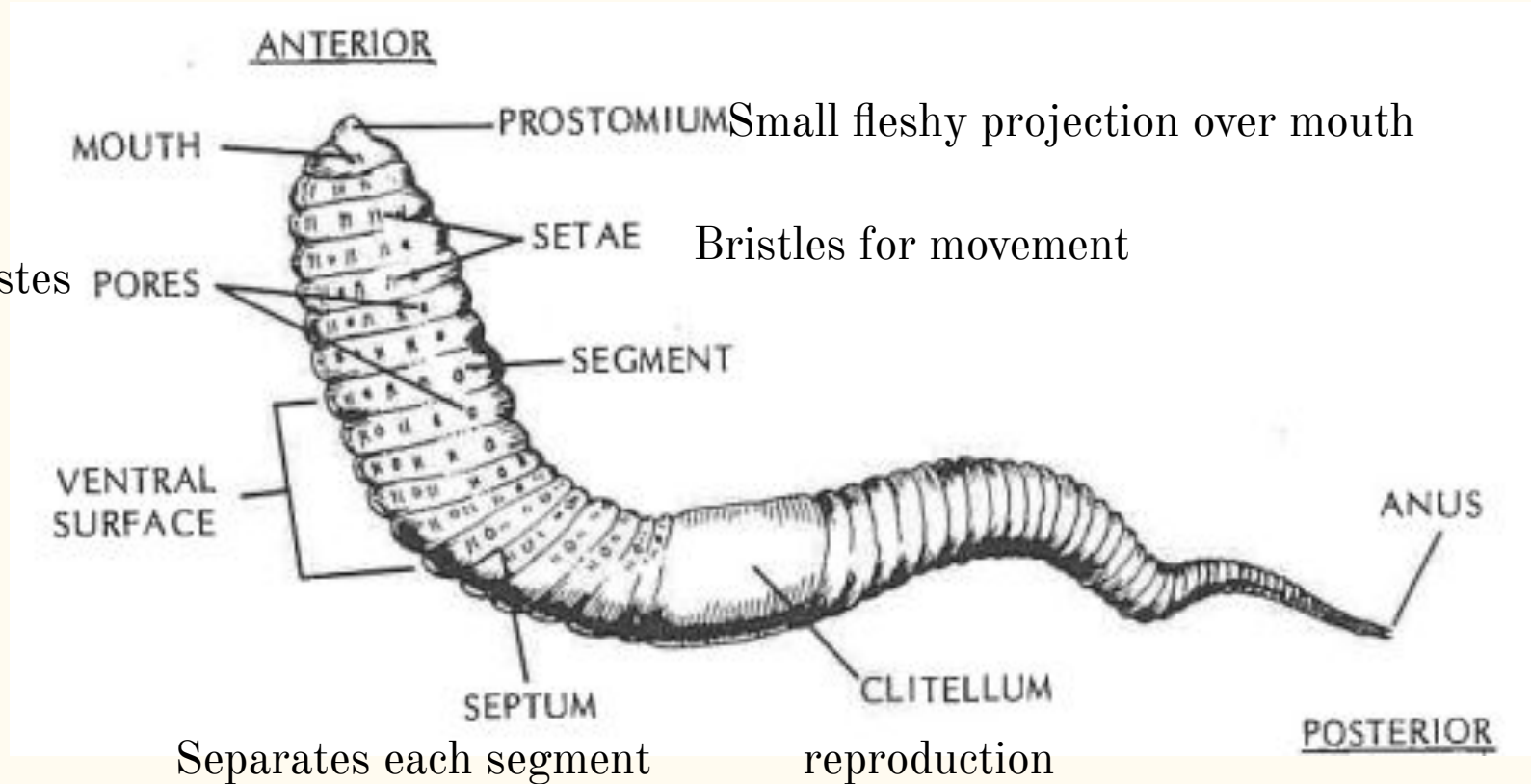
Animal Specific

Earthworm

Earthworm External Anatomy

Food intake

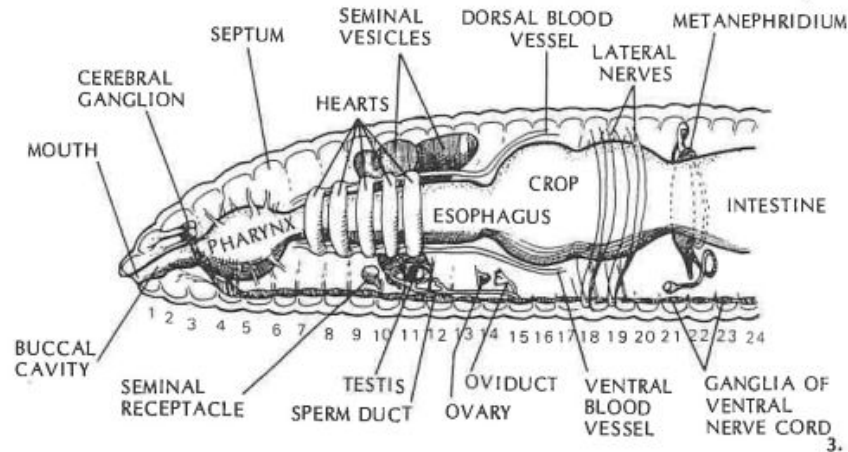
Liquid wastes
excretion



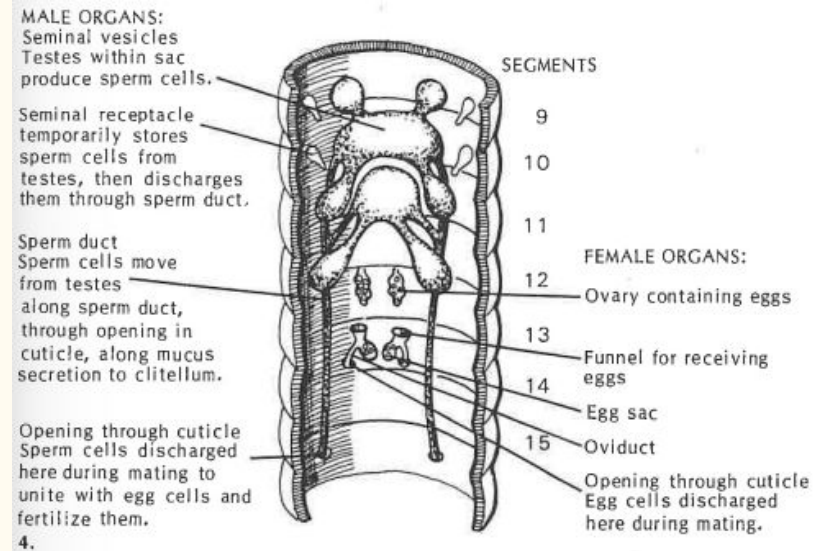
Earthworm Internal Anatomy

Food → mouth → pharynx → esophagus → crop → gizzard → intestine → anus → out

Circulatory system is “closed.” Two major vessels are the dorsal longitudinal and ventral blood vessels. Connected by 5 “hearts” wrapped around the digestive tract as well as other smaller ones



Monoecious Reproductive organs. Seminal vesicles release sperm cells(made by testes) once two worms join together and stored in seminal receptacles of mate.



Grasshopper



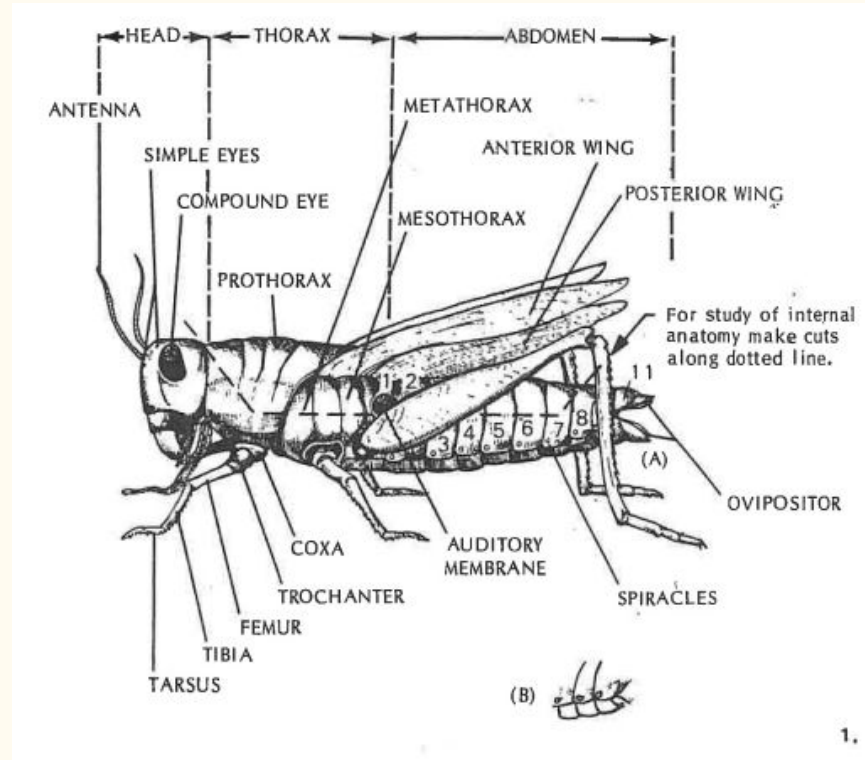
Grasshopper external anatomy

Body is divided into: six segmented head, 3 segmented thorax, to which wings and legs are attached, and an abdomen

Exoskeleton is made almost entirely of chitin which protects internal organs. Occasionally shed this (molting); adults don't molt

Pigmentation provides pattern to match its surroundings

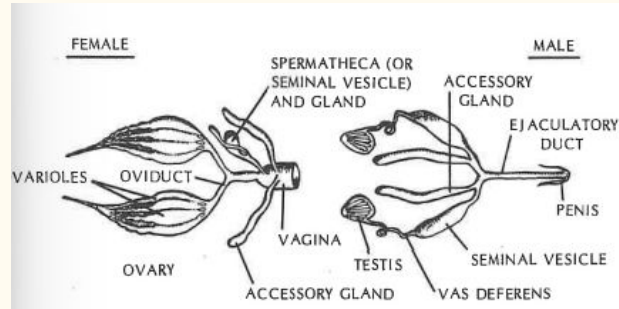
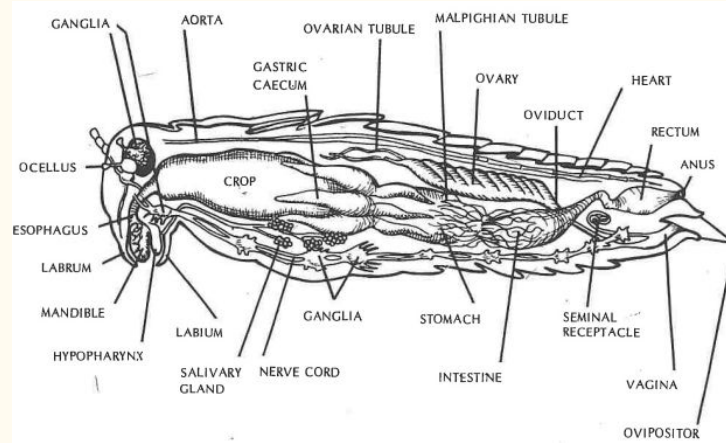
PS: Please study the head of the grasshopper in depth as it is not covered here but may be tested on



Grasshopper internal anatomy

Food → esophagus → crop →
stomach → finger-shaped digestive
glands (gastric caeca) → intestine
→ rectum → anus → out

Numerous tiny tubules, Malpighian
tubules, empty urea and salts from
the blood



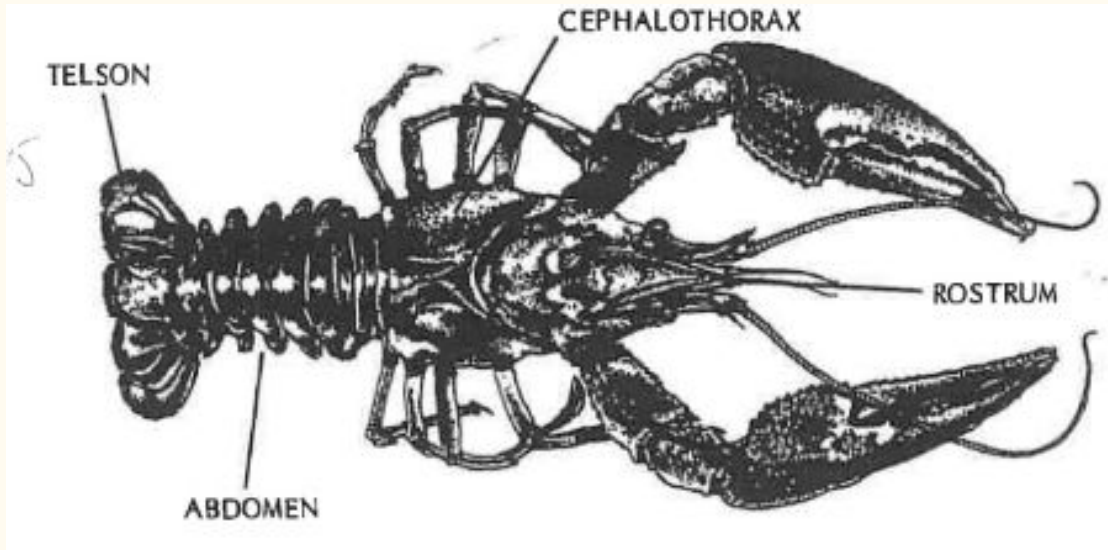
Male - Testis produce sperm, connected
to seminal vesicle, ejaculated through
penis

Female - Vagina collects sperm into
oviduct then transported to ovary

Crawfish

Crawfish External Anatomy

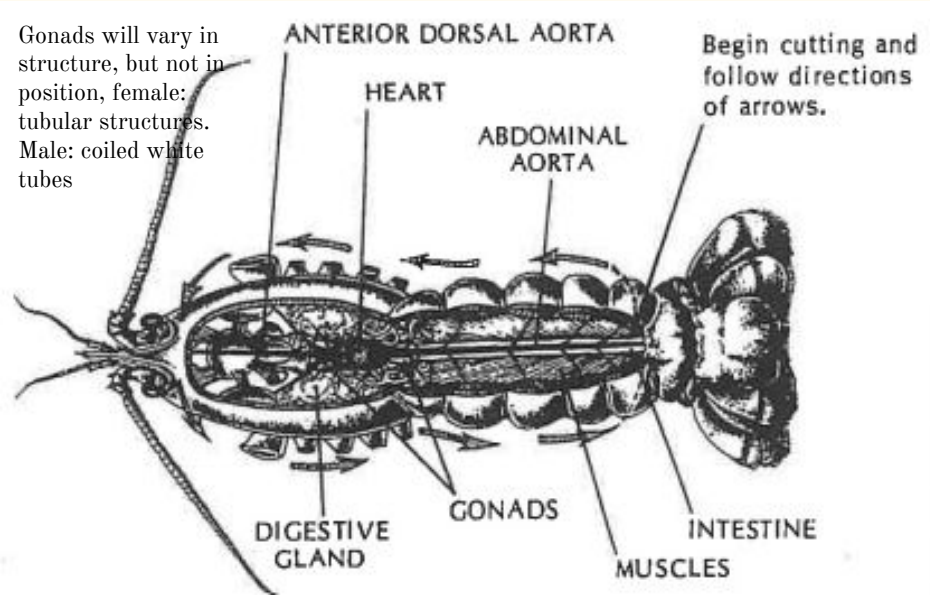
Extension of
final segment



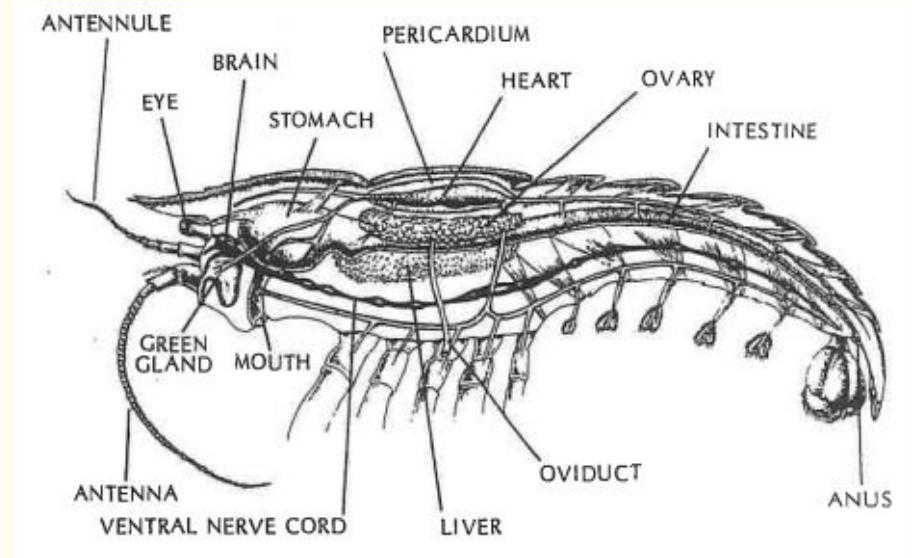
Extension of head in
middle of eyes

The Crawfish contains several different legs and appendages, each serving their own purpose and in a specific location. Please focus on thorough examination of the different appendages on the dissection guides.

Crawfish Internal Anatomy



Blood is transported through vessels and capillaries then into tissue spaces due to the lack of veins.



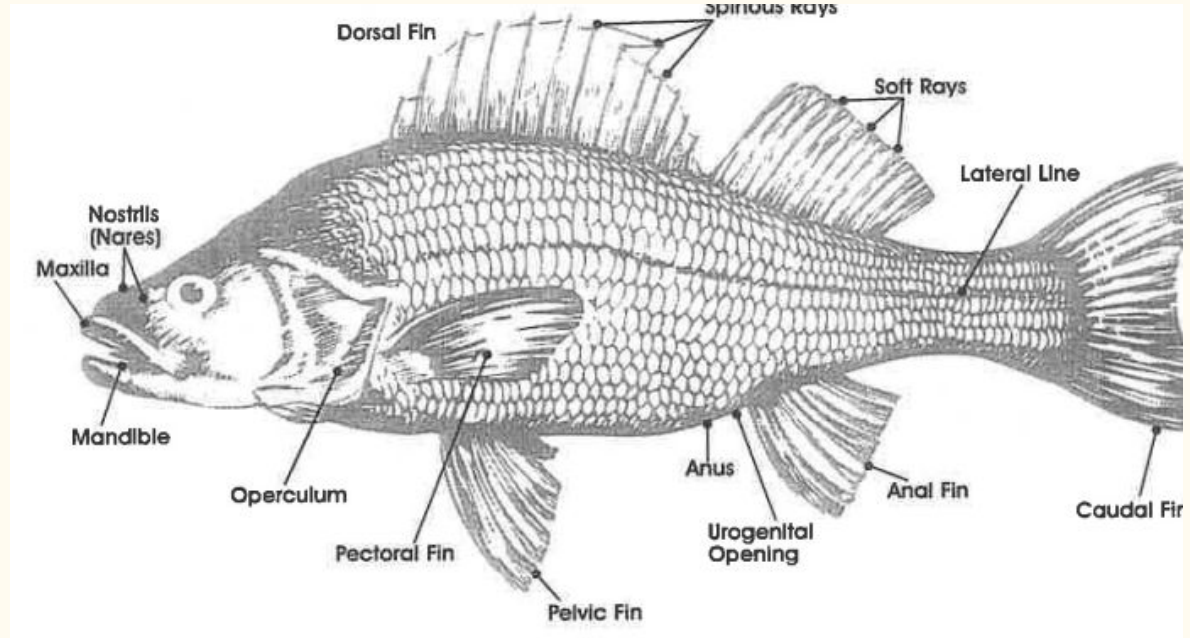
Food → mouth → esophagus → stomach → intestine → anus → out

“Green Glands”: Excretory glands that can also be referred to as the “kidneys”

Fish (Perch)

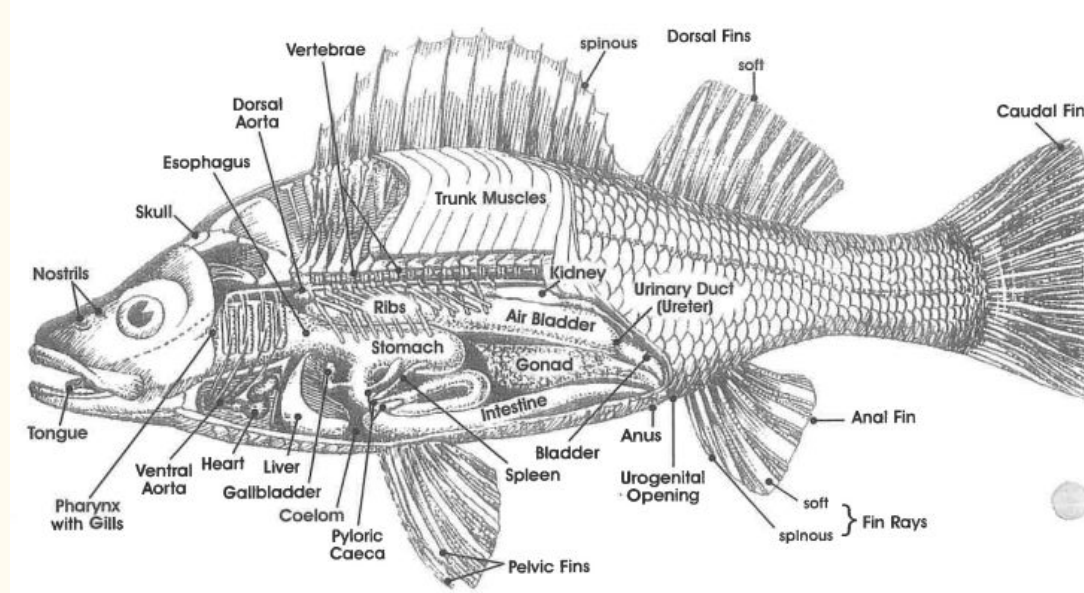
Fish External Anatomy

- Eyes don't have lids
- fins/rays for movement
- Operculum covers gills
- Branchiostegal membrane (lower edge of operculum) is what allows water to pass but doesn't allow it to return



Fish Internal Anatomy

- Food → mouth → esophagus → stomach → intestine → anus
- Air bladder is used to adjust the gravity of the fish
- Waste → kidneys → urinary ducts → urinary bladder → out
- Two chambered heart for single loop circulatory system
- Advanced cephalopodic nervous system (brain)

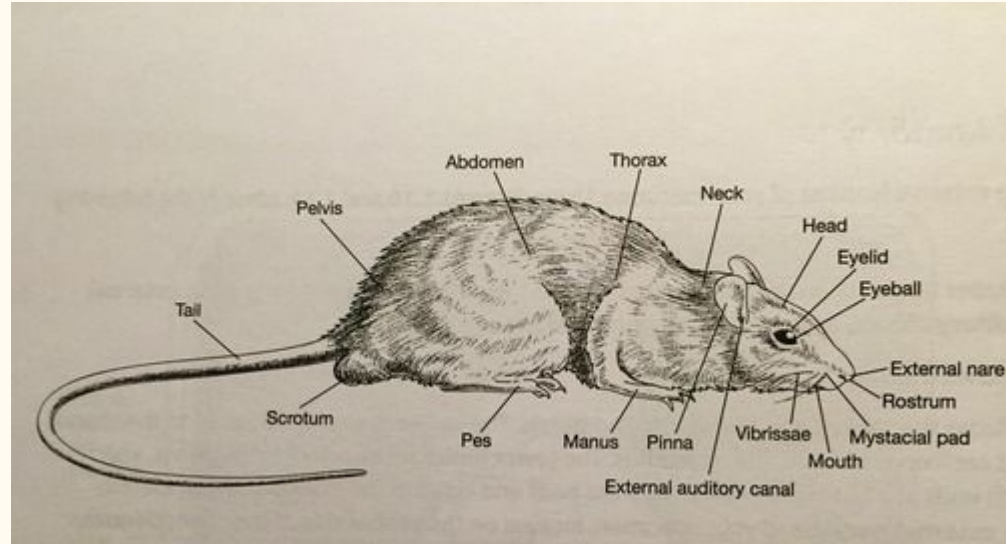


Rat

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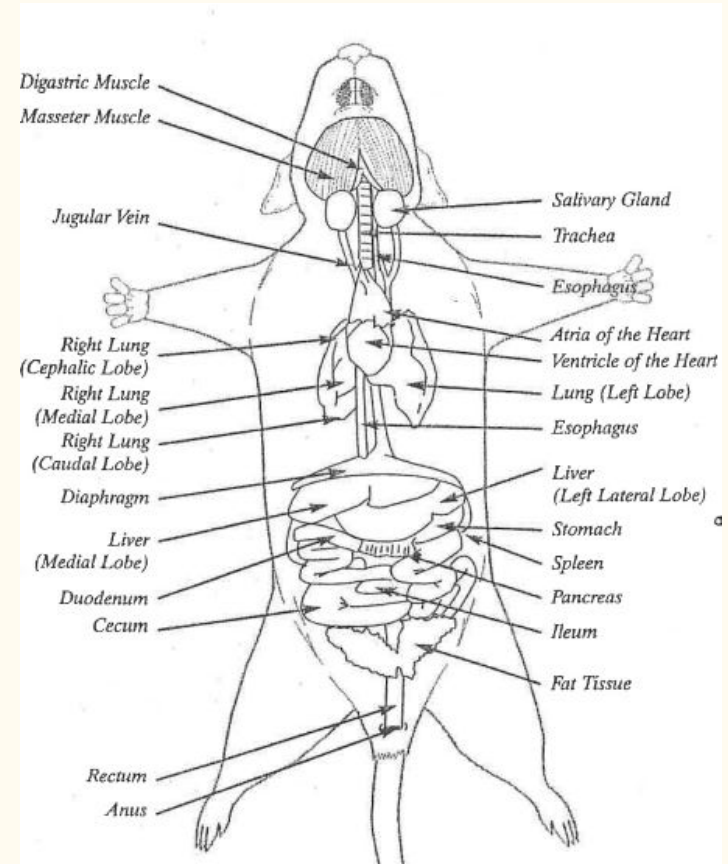
Rat External Anatomy

- Similar traits to that of human
- Tail helps with balance, movement, tracking, etc.
- Above the anus are either the hanging testis or the vaginal orifice
- Nostrils, whiskers, and visible ears
- Vast muscular system



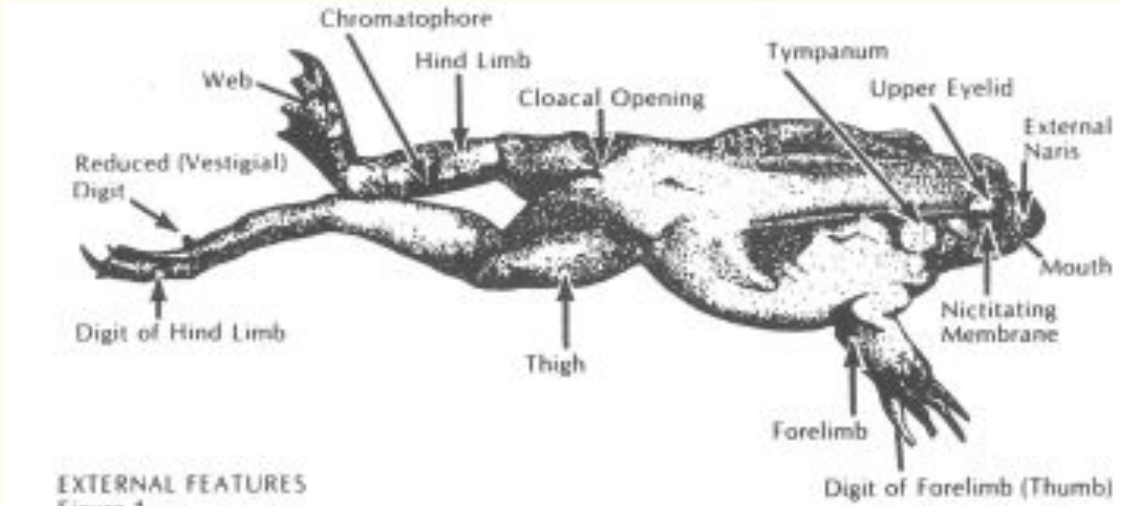
Rat Internal Anatomy

- Double-loop circulation system with four chambered heart (mammal)
- Use lungs and respiratory openings for breathing in Oxygen
- Food → Mouth → Esophagus → Stomach → 3 part intestine → anus → out
- Very advanced anatomy as compared to the prior animals dissected



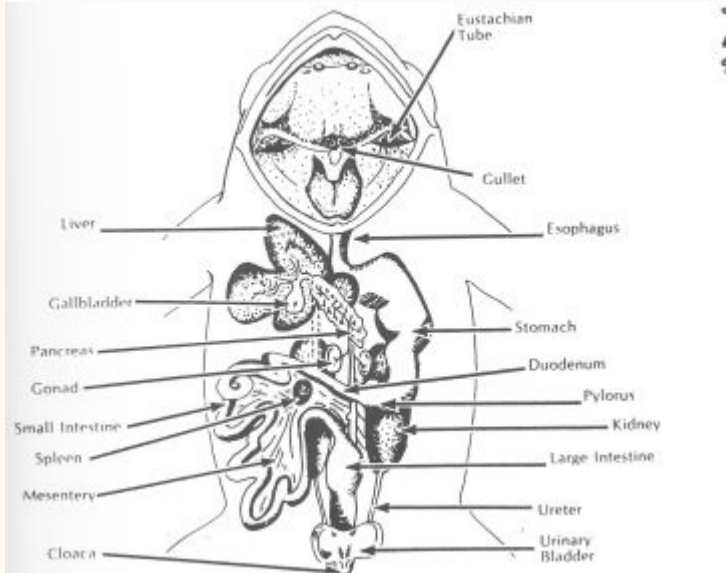
Frog

Frog External Anatomy

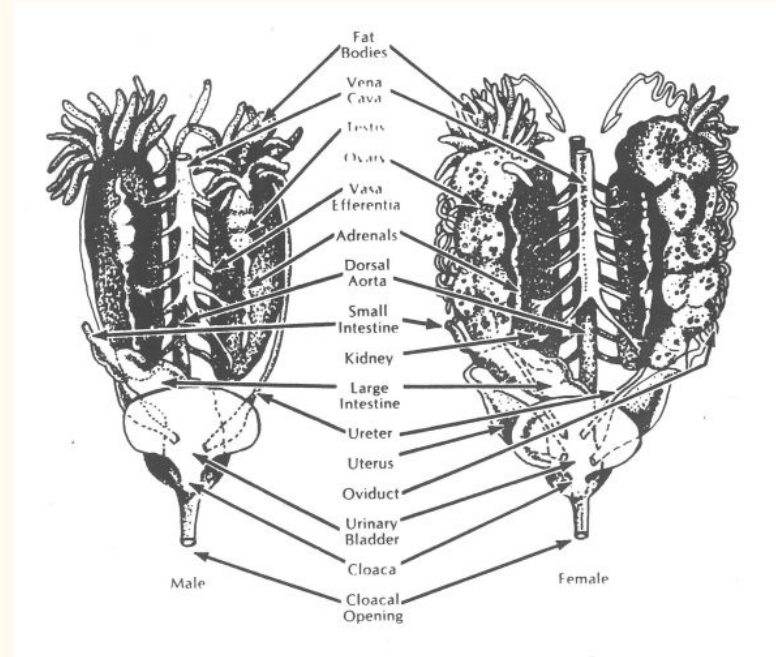


- Three eyelids, including nictitating membrane
- Round membrane covered by skin instead of ears, called tympanum
- Nostrils
- Four limbs

Frog Internal Anatomy



- Advanced digestive system
- Food → Gullet → Esophagus → Stomach → 3 part intestine → anus



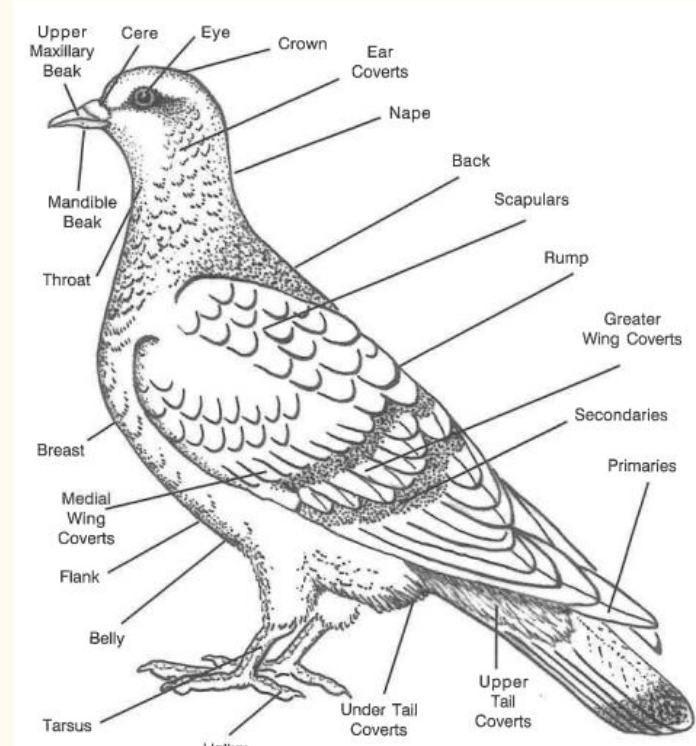
- Nitrogenous wastes excreted from kidney → ureter → urinary bladder
- Double-looped circulatory system

Bird (Pigeon)

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Bird External Anatomy

- Wings for flight
- Feathers are covered in lipid to maintain body heat (water cools the body otherwise)
- Tail for balance and changing direction
- Beak allows for easy access (food)
- Eyes and Ear “Coverts”



Thank you!!

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